Strong-Lensing Time Delays in a Spatially Modulated Closed Cosmology: A Matched Galaxy Benchmark and Conditional Berger–Hopf Response

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Constructed local-encounter checkpoint

Abstract

We compare a conventional galaxy lens with a conditional Berger–Hopf envelope encounter using the same incoming light and physical source screen. Three measured-redshift Jacobi maps in a spatially modulated closed cosmology determine image positions and relative arrival times. For one normalized axial harmonic, six conditional state columns have rank three with nuisances fixed. An analytic factorization explains their rank-zero projection when source, galaxy and mass-sheet freedoms are unrestricted. The isothermal control retains a 17.094532-day relative delay. We also construct a regular, source-free family of constant S^6 target states around a finite Tolman–VII fluid body. The target equations and currents vanish exactly, while the normalized phase stiffness is positive. An explicit common-domain metric/phase Hessian on the Schwarzschild exterior has an invertible metric block and zero mixed response. The solved body produces a nonzero optical encounter, equal to the conventional encounter for the same matter profile; intrinsic mass and radius derivatives are included in image and delay responses. This equivalence is conditional on the retained four-dimensional gravity sector and specified source-free boundary data. A nonzero interface response and the general matching of an exported topographic scalar to the target phase remain open constitutive questions.

Keywords: strong gravitational lensing; time delays; closed cosmology; Jacobi maps; mass-sheet degeneracy; Berger–Hopf framework

1 Physical question and scope

In the Berger–Hopf interpretation, gravitational lensing is the optical response of an envelope event traversed by environmental radiation. The calculation relates prepared incoming light to its outgoing observables, conditional on the envelope state and surrounding geometry. Light is the probe of that state dependence. Traversal alone supplies no claim of encapsulation, de-encapsulation, reflection, absorption or a new photon law.

The quantitative question is whether a change in the encounter geometry or envelope state produces image positions and relative arrival times that survive the available source, galaxy and environmental degeneracies. We retain two explicit realizations: a conventional galaxy-potential benchmark, and a response constructed from the existing Berger–Hopf metric and optical equations. The first executable comparison holds the galaxy encounter fixed and applies the available smooth

geometric correction. We then construct a source-free local target family and a finite gravitating body in a declared effective action sector. Its complete encounter is compared with the same conventional body. A restricted zero response and an exact degeneracy are valid outcomes of the declared comparison.

This focused draft develops the accepted contribution to "Strong Gravitational Lens Modeling," with submission intended by 25 February 2027. It uses the inherited frozen R1 background within its supported domain $0 \le z \le 2.1$. The incident light is a probe of the specified local state; it does not select that state.

2 Matched optical experiment

The observer, lens and source carry orthonormal local tetrads. Two Sachs screen vectors are parallel transported along a common fiducial ray; transverse coordinates are physical lengths in those screens and angles are radians in the local observer or lens frame. We normalize the past-directed affine ray by the observer frequency magnitude $\omega_o = |u_o \cdot k_o| = 1$ and use positive $1 + z_a = \omega_a/\omega_o$. The photon wavelength is assumed much smaller than the metric and encounter scales. Null propagation, not an additional photon interaction law, determines the transfer.

The source emits the same narrow transient marker with the same emission phase, local spectrum, polarization and transverse source coordinate s in both realizations. Its small-amplitude stress is neglected as a test beam. The reference observed central wavelength is 550 nm; at the fixed source redshift the corresponding emission wavelength is 275 nm. The entrance and exit surfaces of the local encounter are planes in the lens rest frame, normal to the fiducial ray, outside the region assigned to the thin galaxy potential. Their thin-plane limit is the lens screen. Propagation outside this encounter uses the smooth reference geometry. The observable is the direction of each received image and the observer proper-time separation between copies of the same marker. Absolute unlensed travel time and a common source-dependent arrival constant are not measured.

The matching prescription fixes the transverse physical source coordinate, the galaxy's physical parameters and the measured lens/source redshifts $z_l = 0.3$ and $z_s = 1.0$. The reference redshift-selected screens consequently move when the metric response changes. At each perturbation images are solved again. Since

$$s = D_{os}\beta, \qquad x = D_{ol}\theta, \tag{1}$$

holding s fixed gives

$$\delta\beta = -D_{os}^{-1}\delta D_{os}\,\beta. \tag{2}$$

Fixing s, β and the image positions independently would describe inconsistent experiments. An alternative fixed inferred angular source is a different protocol and must be labeled accordingly. The longitudinal position and emission event follow each realization's fixed observed-redshift source surface. The physical matching holds the transverse displacement fixed; it does not additionally fix a complete three-dimensional source world-line across the varied geometries.

Cosmological radiation-fluid perturbations (δ_r, v_r) belong to the background state, not to this incoming optical pulse. Muon/reference-state covariances are a third object. Fixing the pulse does not fix its outgoing field or an induced interface response.

3 Coupled state and metric response

The frozen background parameters are

$$(\Omega_{m0}, \Omega_{r0}, \Omega_{k0}) = (0.31, 9 \times 10^{-5}, -0.018), \quad H_0 = 67.4\ \mathrm{km\,s^{-1}Mpc^{-1}}, \tag{3}$$

with retained action parameters $\lambda = 1$, $q = 3$ and $V_0/(M_{\rm Pl}^2 H_0^2) = 2.3986073449708067$. The background is reused without new shooting, calibration or broad stability scans.

For a selected normalized $n = 2$ real spatial harmonic, the physical anchor at $z_i = 2.1$ is

$$Y_i = (\zeta_i, \Pi_{2i}, \delta_{mi}^{\rm ADM}, \delta_{ri}^{\rm ADM}, v_{mi}^{\rm ADM}, v_{ri}^{\rm ADM})^T. \tag{4}$$

The nine-dimensional spatial $n = 2$ eigenspace is not reduced to two physical amplitudes. Selecting one spatial profile leaves six initial amplitudes. The historical two-column seed sets the four ADM fluid coordinates to zero and selects a restricted slice.

For the observer-dipole profile on a reference sphere of radius R_H, one possible normalized choice is

$$Q_{\hat{p}}(\psi, \hat{n}) = \frac{\sqrt{3}}{\pi R_H^{3/2}} \sin(2\psi)\, \hat{n}\cdot\hat{p}, \quad \int_{S^3(R_H)} Q_{\hat{p}}^2 {\rm d}V = 1. \tag{5}$$

The dimensionless unit-sphere convention removes $R_H^{-3/2}$. The selected sky axis $\hat{p}$ is a conditional input, not a direction selected by an isotropic background or by the previous supernova fits. Its radius convention must agree with that of the action provider.

The action-native ADM plus Schutz–Sorkin implementation uses $x = (\zeta, D_m, D_r)$, with $D_m = \delta_m^{\rm ADM} + 3\zeta$ and $D_r = 3\delta_r^{\rm ADM}/4 + 3\zeta$. For auxiliary variables $u = (\alpha, \chi, v_m, v_r)$,

$$\frac{L_2}{a^3} = \frac{1}{2} u^T A u + u^T (B\dot{x} + Dx) + \frac{1}{2}\dot{x}^T K_0 \dot{x} + \frac{1}{2} x^T E x. \tag{6}$$

Constraint elimination gives

$$\begin{aligned} u &= -A^{-1}(B\dot{x} + Dx), & M &= K_0 - B^T A^{-1} B, \\ L &= -B^T A^{-1} D, & V &= E - D^T A^{-1} D, \\ p &= a^3(M\dot{x} + Lx), & \dot{x} &= M^{-1}(a^{-3}p - Lx), \\ \dot{p} &= a^3(L^T\dot{x} + Vx). \end{aligned} \tag{7}$$

These are the coupled original action applications evolved by the provider. The anchor definition $\Pi_2 = 2a_i^3 G_{S,2}(t_i)\dot{\zeta}_i$ is reference-normalized; it is generally not p_ζ. The ADM fluid continuity constraints set $\dot{D}_I = k_n^2(v_I - \chi)$ with the provider's closed-harmonic factor. Thus the reference normalization, reconstructed auxiliaries and Eq. (7) define the anchor-to-canonical map together. No independent momentum identification is imposed.

The reconstructed Newtonian rows use

$${\rm d}s^2 = -(1 + 2\Phi){\rm d}t^2 + a^2(1 - 2\Psi)\gamma_{ij}{\rm d}x^i{\rm d}x^j, \qquad \Phi_W = \frac{\Phi + \Psi}{2}. \tag{8}$$

Potential rows, harmonic normalization and time units are checked before composition. Legacy growth and luminosity-distance modules require a sign and provider adapter; their arrays are not substitutes for the coupled response at this state.

4 Three propagation maps and relative arrival times

For a null tangent and parallel screen, the Jacobi application is

$$\frac{{\rm d}^2 J_a}{{\rm d}\lambda^2} = \mathcal{T}(\lambda) J_a, \quad (J_a, J_a')|_{\lambda_a} = (0, \omega_a I), \quad \mathcal{T}_{AB} = -R_{\mu\nu\rho\sigma} e_A^\mu k^\nu k^\rho e_B^\sigma. \tag{9}$$

Then $D_{ab} = J_a(\lambda_b)$ maps angles in the frame at a to physical separations at b. The three maps D_{ol}, D_{os} and D_{ls} use one reference geometry and compatible frequency/screen normalization. Here J_a already includes the initial factor ω_a; it is not multiplied by that factor a second time. For reverse transport with the same oriented affine derivative, reciprocity reads $D_{ba}/\omega_b = -(D_{ab}/\omega_a)^T$. If the reverse angular opening is instead initialized toward the past endpoint, its derivative has the opposite sign and the positive physical reverse map obeys the corresponding plus-sign relation. The implemented reverse check uses that positive map. In closed FLRW they reduce to

$$\overline{D}_{ab} = a_b S_K(\chi_b - \chi_a) I, \qquad S_K(r) = \frac{\sin(\sqrt{K}r)}{\sqrt{K}}. \tag{10}$$

In particular D_{ls} is independently initialized at the lens, not the difference $D_{os} - D_{ol}$. The main encounter is omitted from all three reference maps when represented as a separate potential.

The existing seven-state reduction is recovered by writing the conformal Jacobi map as $S_K(I + \mathcal{A})$, with $\mathcal{A} = -\kappa I - \left(\begin{smallmatrix} \gamma_1 & \gamma_2 \\ \gamma_2 & -\gamma_1 \end{smallmatrix}\right)$. The directly evaluated sources give

$$Z' = \mathcal{Z}, \qquad \kappa'' + 2\frac{C_K}{S_K}\kappa' = \mathcal{F}, \qquad \gamma_A'' + 2\frac{C_K}{S_K}\gamma_A' = \mathcal{G}_A, \tag{11}$$

where $C_K = S_K'$, $\mathcal{F} = -\operatorname{tr}\delta\mathcal{T}/2$, $\mathcal{G}_1 = -(\delta\mathcal{T}_{11} - \delta\mathcal{T}_{22})/2$ and $\mathcal{G}_2 = -\delta\mathcal{T}_{12}$ in the rescaled conformal system. These fixed-coordinate path variables are passed through Eq. (18) before asserting a fixed measured-redshift response. They are not themselves a complete observed-redshift arrival kernel.

For the declared axial profile, the executed transport differentiates the original angular geodesic equation as an independent construction of the screen maps. Put $q(\chi) = Q_{\hat{p}}(\chi/R_H, \hat{p})$, $P_g = 1{+}2\Phi q$, $M_g = 1 - 2\Psi q$, $A = a\sqrt{P_g}$ and $B_g = a\sqrt{M_g}$. In conformal coordinates $\mathrm{d}s^2 = -A^2\mathrm{d}\eta^2 + B_g^2(\mathrm{d}\chi^2 + S_K^2\mathrm{d}\Omega^2)$. With $N = \ln a$ and static-tetrad frequency E, the equations evaluated are

$$\begin{aligned}
N_{,\chi} &= -aH\sqrt{M_g/P_g}, \\
(\ln E)_{,\chi} &= aH\sqrt{M_g/P_g} - \frac{aH\Psi_{,N}q}{\sqrt{M_g P_g}} - \frac{\Phi q_{,\chi}}{P_g}, \qquad \lambda_{,\chi} = B_g/E, \\
\vartheta_{,\chi} &= \frac{L_\vartheta}{B_g S_K^2 E}, \qquad (L_\vartheta)_{,\chi} = EB_g a^2 q\left(\frac{\Phi}{A^2} + \frac{\Psi}{B_g^2}\right)\vartheta, \qquad D = B_g S_K \vartheta.
\end{aligned} \tag{12}$$

The matter tetrad follows the action-native velocity-potential convention $u_\chi = v_m q_{,\chi}$, rather than an unadapted legacy velocity row. For $s_u = u_\chi/B_g$ and $\gamma_u = \sqrt{1 + s_u^2}$, $\omega = E(\gamma_u + s_u)$. Observed-redshift endpoints are roots of $\omega = 1 + z$. Observer initialization includes the corresponding aberration $\vartheta_o = \gamma_{uo} + s_{uo}$; at the moving lens, $\vartheta_l = 0$ and $L_{\vartheta l} = B_{gl} S_{Kl} \omega_l$. Both screen maps coincide on this axis by rotation symmetry. In particular the angular forcing term in Eq. (12) is retained: suppressing it because the selected profile is a dipole would miss its focusing and endpoint response. Finite perturbations use this explicitly declared truncated metric only to check first derivatives; they are not nonlinear cosmological solutions.

A separate conformal Jacobi control applies the inherited seven-state equation to all six anchor columns, with axial shear zero and forcing

$$\mathcal{F} = q(\Psi'' - 8K\Psi) + 2\Psi'(q' + H_K q) + (\Psi - \Phi)\frac{q}{S_K^2}, \qquad H_K = C_K/S_K. \tag{13}$$

Primes here differentiate along the background χ ray. This is the stable harmonic form of $(\Psi q)'' + 2H_K(\Psi q)' - q(\Phi + \Psi)/S_K^2$; it retains the angular geodesic term. Its observer initialization includes

the matter-tetrad aberration. The measured-redshift endpoint response is then $\delta \ln D_{ob} = \delta N_b + H_{Kb}\delta\chi_b - \kappa_b$, which is compared with the independently varied physical maps for $b = l, s$. The ζ_γ slot is unused and zero in this conformal control; the complete physical frequency and endpoint motion come from the metric ray of Eq. (12). A zero control slot therefore carries no assertion of a zero redshift response.

With $\hat{\alpha} = \nabla_x \hat{\psi}$ in the lens frame, the matrix lens and arrival-time equations are

$$\beta = \theta - D_{os}^{-1} D_{ls} \hat{\alpha}(D_{ol}\theta), \tag{14}$$

$$T = (1 + z_l) D_{ol}^T D_{ls}^{-1} D_{os}, \tag{15}$$

$$ct(\theta, \beta) = \frac{1}{2}(\theta - \beta)^T T (\theta - \beta) - (1 + z_l)\hat{\psi}(D_{ol}\theta) + f(\beta). \tag{16}$$

This is the physical angle-to-separation normalization of the arbitrary reference-spacetime formalism of Fleury et al. [1, Eqs. (2), (3), (11)–(14)]. Their frequency-weighted affine maps become the D_{ab} used here. Reciprocity implies symmetry of T when all maps come from the same reciprocal transport. The numerical antisymmetry residual is inspected before any arrival calculation. Arbitrary symmetrization would conceal an incompatible propagation input. Only differences $t_A - t_B$ are used, so $f(\beta)$ cancels.

4.1 Measured redshifts and moving initialization

Distinguish the fixed-affine and fixed-radial-coordinate frequency variations,

$$Z_{\lambda,a} = \delta \ln \omega_a|_\lambda, \qquad Z_{\chi,a} = \delta \ln \omega_a|_\chi, \qquad Z_{\chi,a} = Z_{\lambda,a} - \mathcal{H}_a \delta\chi_{\mathrm{ray},a}|_\lambda, \tag{17}$$

where $\delta\chi_{\mathrm{ray},a}|_\lambda$ is the ray-coordinate displacement at fixed affine parameter and $d \ln \overline{\omega}/d\chi = \mathcal{H}$. Holding the measured redshift fixed moves the event by

$$\begin{aligned} \delta\lambda_a &= -\frac{Z_{\lambda,a}}{(\ln \overline{\omega})'_a}, \\ \delta\chi_a &= \delta\chi_{\mathrm{ray},a}|_\lambda + \overline{\chi}'_{\lambda,a}\delta\lambda_a \\ &= \delta\chi_{\mathrm{ray},a}|_\lambda - \frac{Z_{\lambda,a}}{\mathcal{H}_a} = -\frac{Z_{\chi,a}}{\mathcal{H}_a}. \end{aligned} \tag{18}$$

Thus $\overline{\chi}'_{\lambda,a}\delta\lambda_a$ alone is a converted affine shift, not the actual radial event displacement. For example, the saved unit-ζ_i source responses are $\partial_{\zeta_i}\chi_s = -0.005611965495588844$ and $\partial_{\zeta_i}\lambda_s = 0.00002815169979075716$ in the declared c/H_0 normalization. At $z_s = 1$, $\overline{\chi}'_\lambda = 4$, so the converted affine shift is $+0.00011260679916302863$. Observer tetrad, source/lens velocity and gravitational frequency terms belong in the frequency variation for the chosen state. The map variation obeys

$$(\delta J_a)'' = \mathcal{T}_0 \delta J_a + \delta\mathcal{T}\, J_{a0}, \quad \delta J_a|_a = -\omega_a I \delta\lambda_a, \quad (\delta J_a)'|_a = \Delta\omega_a I. \tag{19}$$

Here $\Delta\omega_a = \delta\omega_a|_\lambda + \overline{\omega}'_a\delta\lambda_a$ is the total frequency variation at the moving vertex, and vanishes at fixed measured redshift with the observer normalization held fixed. At its output one also adds $J'_{a0}(\lambda_b)\delta\lambda_b$ and the required screen/frame conversion. The zero of J_{a0} at its initial point removes the apparent $J''_{a0}\delta\lambda_a$ term there. Equation (19) is essential for D_{ls}; retaining only a fixed-affine Green integral does not give the measured-redshift observable. A frequency normalization expressed in physical angles must move with the initialization as well.

4.2 Complete first variation

Write $q = \theta - \beta$ and permit a physical encounter change $\delta\hat{\psi}|_x$. At a stationary image, varying Eq. (16) gives

$$\begin{aligned} c\delta t = {} & \frac{1}{2} q^T \delta T q - q^T T \delta\beta - \delta z_l \hat{\psi}(x) \\ & - (1+z_l)\left[\hat{\alpha}(x)^T \delta D_{ol}\theta + \delta\hat{\psi}|_x\right] + \delta f(\beta), \\ \delta T = {} & \frac{\delta z_l}{1+z_l} T + (1+z_l)[\delta D_{ol}^T D_{ls}^{-1} D_{os} + D_{ol}^T D_{ls}^{-1}\delta D_{os} \\ & - D_{ol}^T D_{ls}^{-1}\delta D_{ls} D_{ls}^{-1} D_{os}]. \end{aligned} \tag{20}$$

The image-position derivative cancels by stationarity; the source-coordinate and physical lens-impact terms do not. For the matched fixed-redshift protocol $\delta z_l = 0$, while endpoint shifts remain inside the map changes. If a local envelope potential has not been derived, its $\delta\hat{\psi}|_x$ is unresolved, rather than implicitly supplied by a smooth cosmological response. The executed correction holds the physical conventional encounter fixed and sets this term to zero by the definition of that restricted comparison.

5 Conventional encounter and analytic control

The numerical realization uses a singular isothermal sphere (SIS), with one-dimensional velocity dispersion $\sigma_v = 220\,\mathrm{km\,s^{-1}}$:

$$\hat{\psi}(x) = b_{\mathrm{phys}}|x|, \quad b_{\mathrm{phys}} = 4\pi\frac{\sigma_v^2}{c^2}, \quad \theta_E = b_{\mathrm{phys}}\frac{D_{ls}}{D_{os}}. \tag{21}$$

Here $\hat{\psi}$ has dimensions of length and b_{phys} is a dimensionless deflection amplitude. For scalar distances and $0 < \beta < \theta_E$, the two images and positive saddle-minus-minimum delay are

$$\theta_\pm = \beta \pm \theta_E, \qquad \Delta t_{-+} = \frac{2D_{\Delta t}}{c}\theta_E\beta, \quad D_{\Delta t} = (1+z_l)\frac{D_{ol}D_{os}}{D_{ls}}. \tag{22}$$

The source is away from the caustic and the central singularity. Its baseline position is $\beta_0 = (0.2\theta_{E0}, 0)$ and its held-fixed physical screen coordinate is $s = D_{os,0}\beta_0$. A smooth encounter control uses the same strength in $\hat{\psi}(x) = b_{\mathrm{phys}}\sqrt{|x|^2 + r_c^2}$ with $r_c = 0.05\,\mathrm{kpc} = 5\times10^{-5}\,\mathrm{Mpc}$. Its central image, when present, is retained in the image census rather than discarded to imitate the singular two-image case. The SIS is an illustrative conventional galaxy model, not an inferred mass distribution or a fit to observations. A local encounter specified by Eq. (21) is counted once, outside the reference Jacobi maps.

6 Envelope construction and the available channel

The inherited constitutive chain is

$$\begin{aligned} \mathcal{L}_\psi\psi &= J_{\mathrm{grav}} + \mathcal{E}_\Sigma[Q_\Sigma] + \xi, & h_{\mathrm{topo}} &= \mathcal{R}_{g\psi}\psi, \\ \mathcal{R}_{g\psi} &= -\mathcal{A}_g^{-1}\mathcal{B}_{g\psi}, & Q_\Sigma &= \frac{c^2}{8\pi}\int_{\Sigma_\star} H_\Sigma \Delta K_{uu}\mathrm{d}A. \end{aligned} \tag{23}$$

The inverse and projections require the same physical constraint domain, boundary conditions and state. In the retained static spatial limit, $\mathcal{L}_\psi$ reproduces $-\nabla^2 + B\nabla^4$. The static seam export has the inherited radial shape

$$\mathcal{E}_\Sigma^{(0)}[Q_\Sigma] = \epsilon_\Sigma \frac{Q_\Sigma}{Q} \frac{\Theta(r - r_\star)}{r^2}. \tag{24}$$

Its common-action normalization and orientation remain unresolved. The time derivative of this source for a moving seam is a proposed kinematic continuation; it is not used as an action-selected optical interaction in this checkpoint.

For normalized physical metric basis columns $P_g = (e_A, e_\psi)$ and a normalized local topographic profile φ, define

$$A_g^{\mathrm{loc}} = P_g^\dagger \widehat{H}_{gg} P_g = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad B_{g\psi}^{\mathrm{loc}} = P_g^\dagger \widehat{H}_{g\varphi}\varphi = \begin{pmatrix} u \\ v \end{pmatrix}. \tag{25}$$

The inverse application and local Schur curvature are

$$\begin{aligned} \mathcal{R}_{g\psi}^{\mathrm{loc}} &= \frac{1}{ac - b^2} \begin{pmatrix} bv - cu \\ bu - av \end{pmatrix}, & \Delta_g &= ac - b^2 \neq 0, \\ k_{\mathrm{loc}} &= c_\varphi - (B_{g\psi}^{\mathrm{loc}})^T (A_g^{\mathrm{loc}})^{-1} B_{g\psi}^{\mathrm{loc}}, & c_\varphi &= \langle \varphi, \widehat{H}_{\varphi\varphi}\varphi \rangle. \end{aligned} \tag{26}$$

The diagnostic is $\|A_g^{\mathrm{loc}} \mathcal{R}_{g\psi}^{\mathrm{loc}} + B_{g\psi}^{\mathrm{loc}}\|$, with the determinant, basis norms, Legendre gate and boundary-domain status reported. These coefficients are action projections, not free fitting parameters. The cosmological lapse/shift auxiliary Hessian in Eq. (6) does not automatically supply this local physical metric/topographic Hessian.

The available same-domain action projection is nevertheless executed on the ADM auxiliary block at $z_l = 0.3$. After eliminating the fluid velocity auxiliaries its lapse/shift entries, in the action-density convention L_2/a^3 and unit normalized harmonic coordinates, are

$$(a_{\mathrm{ADM}}, b_{\mathrm{ADM}}, c_{\mathrm{ADM}}) = (-7.96725028,\ 0.56954480,\ 0.48267978), \qquad \det A_{\mathrm{ADM}} = -4.17001193. \tag{27}$$

All six mixed source columns are evaluated; their inverse response has residual 1.42×10^{-15}, the original action constraints have maximum residual 5.28×10^{-16} and the Schur reduction agrees with the native reduced action to 4.16×10^{-16}. The unit ζ source has pre-elimination self entry 0.301115412 and Schur entry 5.097094780. The reduced kinetic eigenvalues are $(0.00418211, 0.17251064, 0.43386354)$, so its Legendre gate is satisfied at this event. These data preserve the common cosmological domain and demonstrate its actual auxiliary inverse application. The map $\Phi = \alpha + \dot{\chi}$, $\Psi = -\zeta - H\chi$ depends on state and time derivatives, so it cannot rename this block as the local physical A_g^{loc} of Eq. (25).

In the established static block-diagonal spatial phase channel the mixed source is explicitly

$$\begin{aligned} \mathcal{S}_{ij}[j, \varphi] &= 2F''(X)(j^k \nabla_k \varphi) A_{ij} + 2F'(X) j_{(i} \nabla_{j)} \varphi - F'(X) g_{ij} (j^k \nabla_k \varphi), \\ (u, v) &= \int \mathrm{d}\mu_g \, w \, ((e_A)^{ij}, (e_\psi)^{ij}) \mathcal{S}_{ij}. \end{aligned} \tag{28}$$

For $j_i = 0$, or for $\nabla_i \varphi = 0$, the pointwise source and both projections vanish exactly. On any invertible admissible metric block this gives $\mathcal{R}_{g\psi}^{\mathrm{loc}} = 0$ and $k_{\mathrm{loc}} = c_\varphi$. This is a channel-specific null result; it does not determine a nonzero-current state or prove that all lensing vanishes. No new channel is introduced to manufacture a signal. For a spatially constant *phase variation*, this derivative sector has zero self entry. A constant reference texture does not imply zero self entry for a nonconstant phase variation: the construction below evaluates its positive Dirichlet stiffness.

The smooth coupled cosmological metric response is separately available from Eqs. (6)–(8). Its transfer per anchor amplitude is a conditional state sensitivity. Until a physical envelope state and the common-domain local coefficients are supplied, that sensitivity is neither a galaxy-replacement amplitude nor a calibrated BHSM prediction.

7 Nuisance projection and distinguishability

Let the local observable vector contain the image coordinates and independent relative delays. In the linearized model

$$y = BX + N\eta + \epsilon, \qquad WCW^T = I, \quad P = I - (WN)(WN)^+, \quad F = (WB)^T P(WB), \tag{29}$$

B contains conditional physical-state columns, while N contains source coordinates, galaxy parameters, mass-sheet scale, independent line-of-sight contributions and cosmological nuisance columns admitted by the declared protocol. The covariance C is a specified illustrative measurement covariance, not an estimated observational likelihood. Singular values of PWB report the surviving rank and parameter-space null directions. On a null direction there is no likelihood-derived finite uncertainty. The zero eigenvalue of a Fisher pseudoinverse must not be reported as a zero error bar. A prior must be added and identified before it can supply missing information.

For a reduced scalar lens the mass-sheet transformation is

$$\psi_\lambda(\theta) = \lambda\psi(\theta) + \frac{1-\lambda}{2}|\theta|^2, \quad \beta_\lambda = \lambda\beta, \quad \Delta\phi_\lambda = \lambda\Delta\phi. \tag{30}$$

It preserves image positions under the corresponding source rescaling. An independent unrestricted delay scale for each lens absorbs a common rescaling of all its delays. More sky directions alone do not remove that freedom. Delay ratios can remove a common scale only where multiple independent delays exist; a two-image SIS has just one independent delay. Image distortions or independent environmental/kinematic data require their own response and likelihood.

The dominant-lens treatment permits subcritical line-of-sight corrections beyond a single external convergence/shear [2]. In particular observer–lens, observer–source and lens–source segments enter differently; treating only the observer–source convergence can bias cosmographic inference [3]. Their physical contributions are already contained in the three reference maps here and are not added again as a second convergence potential. An independent nuisance column represents uncertainty in an included contribution, not another copy of its mean.

The familiar H_0^{-1} delay scaling follows when dimensionless expansion history, reduced angular galaxy parameters and angular source coordinates are fixed: every distance scales as H_0^{-1}. Holding the physical source coordinate fixed instead changes β, and therefore requires the full variation in Eq. (20). The two conventions are not interchangeable. In the scale-free SIS control, the fixed physical transverse source gives the exact identity

$$\Delta t_{-+} = \frac{2(1+z_l)b_{\mathrm{phys}}s}{c}\frac{D_{ol}}{D_{os}}, \qquad s = |X_s|. \tag{31}$$

It is independent of a common H_0^{-1} distance rescaling when the dimensionless expansion/curvature history, measured redshifts, b_{phys} and s are fixed and the two-image branch is retained. This cancellation does not apply to arbitrary changes of geometry or to a family with another fixed dimensional lens scale. The independently checked fixed angular-source convention instead gives $\partial\Delta t/\partial \ln H_0 = -\Delta t$.

7.1 Analytic scalar-map nuisance degeneracy

For the centered softened-isothermal family used below, put

$$\beta = \frac{s}{D_{os}}, \quad \theta_E = b\frac{D_{ls}}{D_{os}}, \quad \theta_c = \frac{r_c}{D_{ol}}, \quad T = (1+z_l)\frac{D_{ol}D_{os}}{D_{ls}}. \tag{32}$$

Here $b = b_{\rm phys}$ and r_c is a physical core radius; the SIS is $r_c = 0$. The reduced potential is $\psi_{\rm red}(\theta) = \theta_E\sqrt{|\theta|^2+\theta_c^2}$. Images depend only on $(\beta, \theta_E, \theta_c)$, and relative delays are T/c times differences of its reduced Fermat potential.

Proposition (scalar-map controls). At fixed measured lens redshift and fixed physical (s, b, r_c), let $f_{ab,a} = \partial_a \ln D_{ab}$. For noncaustic images on fixed branches, the state-response matrix of image coordinates and relative delays obeys

$$B = N_{\rm min}R, \qquad N_{\rm min} = [N_{{\rm source}\,x}\ N_{\ln b}\ N_{\ln r_c}\ N_{\rm MST}],$$
$$R = \begin{pmatrix} -\beta_{\rm as} f_{os} \\ f_{ls} - f_{os} \\ -f_{ol} \\ f_{ol} + f_{os} - f_{ls} \end{pmatrix}. \tag{33}$$

Each entry of R is a row over state coordinates; $\beta_{\rm as}$ is in arcseconds because the implemented source nuisance is per arcsecond. Delete the core-radius column and its row for the SIS. The implemented compensated mass-sheet column leaves images fixed and rescales all relative delays. It therefore supplies the $\delta \ln T$ response.

To prove the identity, differentiate Eq. (32): $\delta\beta_{\rm as} = -\beta_{\rm as} f_{os}\delta Y$, $\delta \ln \theta_E = (f_{ls} - f_{os})\delta Y$, $\delta \ln \theta_c = -f_{ol}\delta Y$, and $\delta \ln T = (f_{ol} + f_{os} - f_{ls})\delta Y$. At fixed maps the first three nuisance columns generate precisely the first three variations; the delay-scale column generates the last. The chain rule establishes Eq. (33). For any positive-definite observational covariance C, set $W = C^{-1/2}$ and let P project orthogonally off the range of $WN_{\rm min}$. Then $PWB = 0$ when these nuisance directions are unrestricted. No additional line-of-sight or H_0 nuisance is needed for this result. The compensating nuisance transformation changes source and galaxy parameters: this is an observable degeneracy, not physical identity of the underlying preparations.

The saved response arrays give relative whitened Frobenius discrepancies

$$\frac{\|W(B - N_{\rm min}R)\|_F}{\|WB\|_F} \tag{34}$$

of 2.45294×10^{-12} (SIS) and 6.37772×10^{-12} (smooth core). These finite-difference residuals support the analytic identity for the specified scalar-map experiment. An intrinsic local encounter or nonscalar propagation requires its own response and nuisance analysis.

A pure sky dipole has zero trace-free angular Hessian at Born order, $(\nabla_a\nabla_b Q)^{\rm TF} = 0$; its shear null does not remove its redshift or isotropic focusing response. Likewise, for m coherent global amplitudes and a fixed response M_O,

$$C_O^{\rm topo} = M_O C_q M_O^T, \qquad \operatorname{rank} C_O^{\rm topo} \le \operatorname{rank} C_q \le m. \tag{35}$$

This bound concerns the isolated coherent contribution. One local random field with independent increments can generate many nonparallel response columns and need not have rank one. No Gaussian factorization of a dressed state or substitution of a retarded kernel for a statistical state kernel is used.

8 Executed numerical checkpoint

The generated table below is tied to the input manifest and machine-readable calculation rather than to a manuscript-only amplitude choice. Conditional perturbations use the same source convention as the benchmark. Centered finite changes re-solve the image equation independently before comparing arrival-time differences with Eq. (20).

Quantity	Executed result
Lens/source redshift	0.3/1.0
D_{ol}, D_{os}, D_{ls} [Mpc]	945.162839, 1689.312240, 1076.994348
SIS image x positions [arcsec]	1.0678831, -0.7119220
SIS relative delay [days]	17.09453205
Free-nuisance surviving state rank	0
Fixed-nuisance control rank	3
Synthetic-prior-assisted rank	3
Static spatial phase (u, v)	$(0, 0)$ under the stated current gate

The physical background maps are

$$(D_{ol}, D_{os}, D_{ls}) = (945.162839, 1689.312240, 1076.994348)\,\mathrm{Mpc}. \tag{36}$$

The source-screen displacement is $(1.457663367, 0)$ kpc. The SIS images are $(1.0678830532, 0)$ and $(-0.7119220355, 0)$ arcsec, with relative delay 17.0945320516 days. Their original lens-equation residual is below 8.5×10^{-22} radians and their arrival stationarity residual is zero at reported floating precision. The analytic scalar SIS delay agrees within 1.1×10^{-14} days. Baseline segment reciprocity residuals are below 9×10^{-16}, and the assembled time-delay matrix is symmetric without modification.

In the anchor order of Eq. (4), the executed fractional relative-delay sensitivities are

$$\begin{aligned}\frac{1}{\Delta t}\frac{\partial \Delta t}{\partial Y_i} = (&-1.16307 \times 10^{-5},\ 2.83592 \times 10^{-3},\ 2.79299 \times 10^{-4},\\ &7.83676 \times 10^{-8},\ -2.69536 \times 10^{-3},\ 5.31244 \times 10^{-7}).\end{aligned} \tag{37}$$

Each entry is per unit of its declared physical anchor coordinate with the normalized spatial profile $N_Q = 0.02709350135$ in $H_0 = 1$ units. The six columns use independent coupled fluid as well as scalar initial conditions; they are not the older restricted two-column calculation.

For the explicitly specified probe $Y_i = (10^{-3}, 0, 0, 0, 0, 0)$, the images shift by $(+4.07393 \times 10^{-8}, -5.56173 \times 10^{-8})$ arcsec along the image axis. The independently solved relative delay changes by -1.98831×10^{-7} days, or -0.0171790 seconds. The first-order prediction is -1.98821×10^{-7} days. On that trial ray, $\max|\Phi| = \max|\Psi| \simeq 1.98 \times 10^{-6}$ and the largest relative metric perturbation is 3.96×10^{-6}. This is a numerical trial of a declared state amplitude, not a realized or calibrated physical envelope state. Its additional local static spatial phase response under the stated zero-current condition is exactly zero.

The complete relative-delay derivative for the unit ζ_i column separates into approximately zero from δT, -7.14118×10^{-4} days from source-coordinate remapping, $+5.15297 \times 10^{-4}$ days from physical galaxy impact and zero from the held-fixed intrinsic encounter. Their sum is -1.98821×10^{-4} days per unit ζ_i. Dropping the source or impact term would give a different response. Across all six columns, independently re-solved half-step image/delay derivatives agree with the complete response to a maximum covariance-whitened relative difference of 1.45×10^{-7}. The independent

conformal seven-state control, including the measured endpoint shifts above, reproduces all six observer–lens and observer–source map derivatives to a maximum absolute difference of $9.72 \times 10^{-11}\, c/H_0$ per anchor unit.

For the declared covariance, image-coordinate errors are 0.001 arcsec and the delay error is 0.1 days, with zero cross-covariances. With all nuisances fixed, the nonzero whitened state-response singular values are $(7.81204, 2.86044, 0.00402428)$: three combinations of the six amplitudes remain null even in this control. The unrestricted source, galaxy, mass-sheet, line-of-sight and physical-source H_0 nuisance matrix spans all five observable rows. The surviving state rank is zero, with largest projected singular value 1.88×10^{-15} against the declared 7.81×10^{-8} rank threshold. The whole six-dimensional state space is therefore unconstrained by this experiment under these nuisance freedoms. Delay-only projection also has rank zero. The synthetic nuisance prior control has rank three; it supplies independent assumed information, not an empirical constraint or detection.

The smooth-core check retains three images, at approximately $(1.067837, -0.002257, -0.711817)$ arcsec. Its central and outer relative delays are 30.259543 and 17.093193 days with respect to the positive outer image, giving ratio 0.56488603. The ratio's mass-sheet derivative vanishes, as required, while its source, strength and core derivatives remain nonzero. Allowing the declared smooth-galaxy and source nuisances still leaves state rank zero. Thus a mass-sheet-invariant ratio does not by itself identify the conditional state in this realization. The central-to-positive-outer magnification ratio is 2.79565×10^{-5}; equal assumed coordinate precision for this faint central image is an idealized control, not an observational forecast.

The first comparison is valid in geometric optics, linear response of the smooth geometry, small angles, and the thin, nonrelativistic conventional galaxy encounter. It excludes caustic crossings under the small finite variations used in the derivative check. An extended envelope would require integrating its derived metric through the entrance/exit surfaces before justifying any thin-plane reduction. This calculation does not supply that reduction by assumption.

9 A constructed source-free local family

9.1 Retained action, fields and conditional data

We now evaluate a local encounter, rather than identify its missing inputs. The retained four-dimensional action is Einstein gravity plus ordinary perfect-fluid matter and the recovered physical Path–B target sector,

$$S_{\rm loc} = \frac{c^4}{16\pi G} \int \mathrm{d}t \mathrm{d}^3x\, N\sqrt{h}\, R + S_{\rm fluid} + S_{\rm GHY} + S_\eta + S_{\rm YM}, \tag{38}$$

$$S_\eta = -\int \mathrm{d}t \mathrm{d}^3x\, N\sqrt{h}\, w(\sigma) F(X) + S_{\rm unit}, \tag{39}$$

$$F(X) = \frac{\kappa_\eta}{2} X + \frac{1}{8} X^4, \qquad X = g^{\mu\nu} \langle D_\mu \eta, D_\nu \eta \rangle. \tag{40}$$

The target is $S^6 \subset \mathbb{R} \oplus \mathbb{C}^3$, with $\eta = (s, z)$, $s^2 + 2z^\dagger z = 1$ and target norm $\mathrm{d}s^2 + 2\mathrm{d}z^\dagger \mathrm{d}z$. The connection is $SU(3)$ on z; N is the physical static lapse and h the spatial metric. Here κ_η denotes the original target coefficient κ_1, distinguished from the gravitational coefficient. We require $w_0 \kappa_\eta > 0$; its dimensional value remains a specified coupling, not a lens fit. The higher-power coefficient and the dimensions of X are those of the original target-sector units. The intrinsic boundary action contains $C_\partial R_4$; setting $C_\partial = c^4/(16\pi G)$ defines the retained effective gravity sector. This does not derive the cap reduction or its reaction terms. We condition on no additional local cap traction or vacuum energy in this sector. Restoring a nonzero cap reaction requires its own field equation.

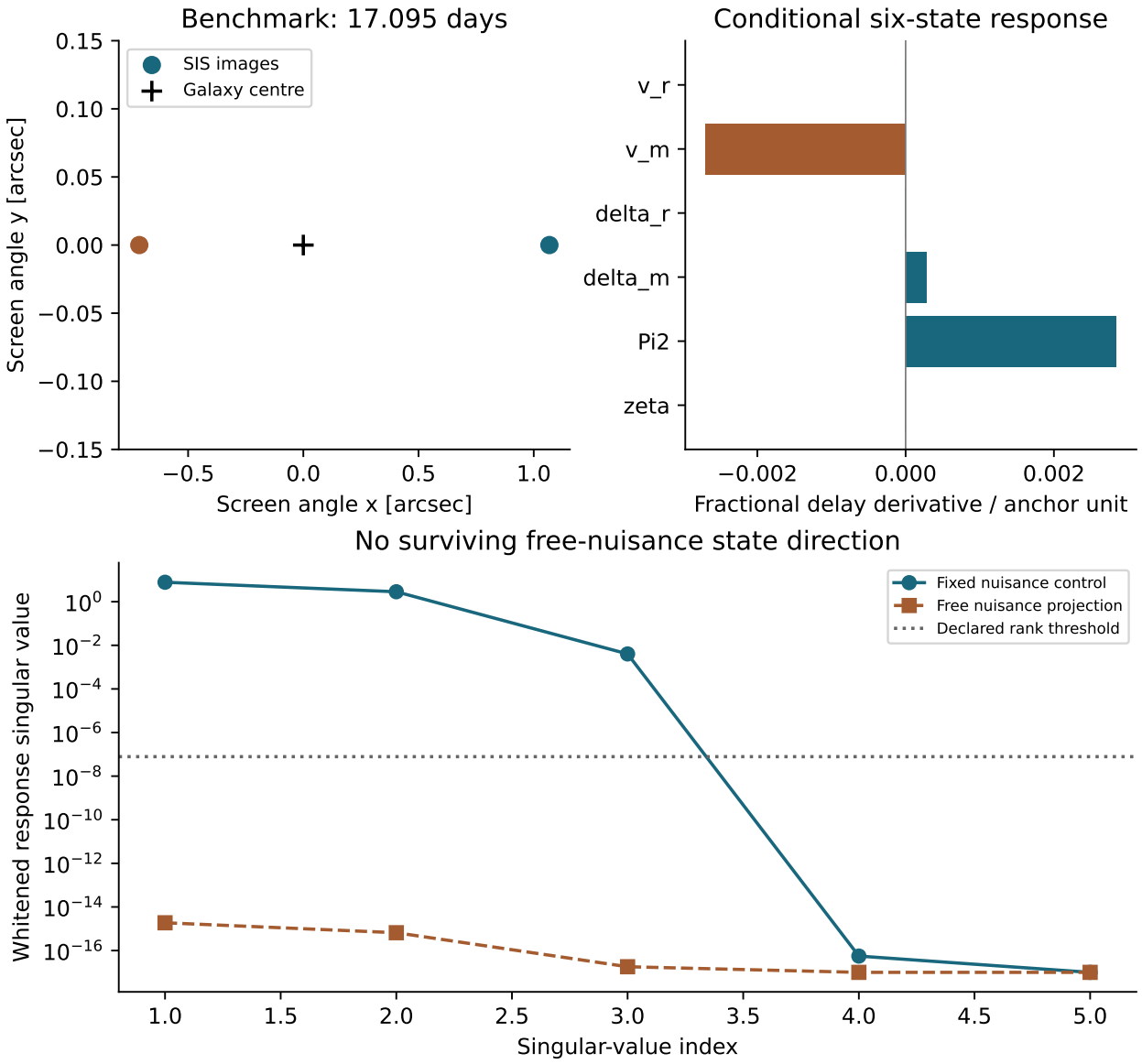


Figure 1: Executed illustrative comparison and conditional sensitivities. The input manifest defines the perturbation, covariance and source conventions. These are benchmark calculations rather than an observational fit or an action-normalized local envelope prediction.

The local isolated solution is inserted in the declared weak-field comparison with cosmological propagation. Pure background-curvature corrections scale as $(HR/c)^2$. A joint solution with a rolling scalar or cap reaction is not supplied, and that estimate does not bound such mixed responses. This is a conditional local effective sector, not a complete stratified or globally matched BHSM galaxy solution.

The source equations and their version-pinned implementation are recovered from the Path–B foundational action v14.31, physical topology v14.32, metric/phase source v14.39 and Lorentzian Legendre gate v14.87. The input receipt records their repository commit, Git blob and SHA256. The recovered source repository is pinned at `524ed90689bd5923c249bba2e699abf627e703cd`; the focused derivation and eight source hashes accompany the calculation. In particular, the physical $M_4 \to S^6$ sector does not inherit a degree-one seven-dimensional texture charge.

For each specified constant q, choose the real target meridian

$$\eta_q = (\cos q,\, \sin q\, e_1/\sqrt{2}), \qquad A_\mu = 0, \qquad w = w_0 > 0. \tag{41}$$

The parameter $s = \cos q$ is $SU(3)$ invariant, so this is not just a gauge orbit. It is not claimed to be a uniquely selected environment. Let $T\eta = \partial_q \eta$ on this meridian; $|T\eta|^2 = 1$. This tangent is a target-sector direction, not an $SU(3)$ generator annihilating the north pole. All reference derivatives vanish: $D\eta = 0$, $X = F(0) = 0$, $A_{\mu\nu} = \langle D_\mu\eta, D_\nu\eta\rangle = 0$ and $j_\mu = \langle D_\mu\eta, T\eta\rangle = 0$. The target Euler equation, unit multiplier, Yang–Mills current/Gauss row, stress tensor and momentum row therefore vanish exactly, on any metric. The additional σ force is $w'(\sigma)F(0) = 0$. This adds no η-dependent σ/cap source. The independent local σ/cap equations are outside the retained effective truncation.

The constant family also follows from the declared source-free radial problem. For $q = q(r)$, $A_\mu = 0$ and $\mathrm{d}s^2 = -e^{2\nu}c^2\mathrm{d}t^2 + f_r^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$, the original target equation is

$$\partial_r\{r^2 e^\nu \sqrt{f_r}\, w_0(\kappa_\eta + X^3)q'\} = 0, \qquad X = f_r q'^2. \tag{42}$$

Its conserved conormal flux must be zero at a regular centre. Positivity of $\kappa_\eta + X^3$ forces $q' = 0$ everywhere. A prescribed constant Dirichlet boundary value fixes q. A nonzero radial texture would consume an inner boundary charge or external source, absent in this problem.

9.2 Finite body, field equations and matching

The shared conventional and envelope body has specified mass $M = 10^{11} M_\odot$ and areal support $R = 3\,\mathrm{kpc}$, chosen as finite illustrative matter data rather than tuned to a deflection. We use the quadratic-density Tolman–VII profile [4],

$$\rho(r) = \frac{15M}{8\pi R^3}(1 - r^2/R^2), \quad m(r) = \frac{M}{2}(5x^3 - 3x^5), \quad x = r/R \le 1, \tag{43}$$

and vacuum outside. Pressure is solved, not replaced by stationary dust. The applications evaluated are

$$m' = 4\pi r^2 \rho, \qquad f_r = 1 - \frac{2Gm}{c^2 r}, \tag{44}$$

$$\nu' = \frac{G(m + 4\pi r^3 p/c^2)}{c^2 r^2 f_r}, \qquad p' = -(\rho c^2 + p)\nu'. \tag{45}$$

The outer Cauchy data are $p(R) = 0$ and $\nu(R) = \frac{1}{2}\ln(1 - 2GM/(Rc^2))$. Inward integration gives finite central pressure and lapse. Monotone $\rho(r)$ defines a conditional barotropic relation

$p(\rho)$ along the solution; the evaluated $0 \le \mathrm{d}p/\mathrm{d}(\rho c^2) < 1$ check excludes the incompressible-fluid shortcut. This is a specified fluid model, not a stellar-population fit. The original Einstein radial and angular equations and conservation residuals are recorded in the result receipt. At R, $m = M$, $p = \rho = 0$; the induced metric and extrinsic curvature match Schwarzschild with continuous ν'. No Israel surface stress or target conormal flux is required. The body, reference target, source and matching conditions are the same for both optical calculations. For the evaluated body, the Hamiltonian and radial Einstein residuals scaled by $\epsilon = GM/(Rc^2)$ are below 4.3×10^{-15}; the angular residual scaled by ϵ^2 is 2.23×10^{-9}. An independent derivative of the integrated lapse gives a scaled Einstein residual 2.38×10^{-12}. These are equation/discretization checks, not errors on the physical matter model.

9.3 Exact common-domain physical Hessian and source

The local projection uses the vacuum annulus in isotropic radius $r_0 < \varrho < L$, with

$$m_g = GM/c^2, \quad r_0 = \tfrac{1}{2}(R - m_g + \sqrt{R(R-2m_g)}), \quad L = 2R, \quad \ell = L - r_0. \tag{46}$$

This exterior domain avoids an unspecified fluid second variation. Its boundary metric is fixed by the solved body and exterior solution. Write $N = e^{\Phi}$, $h_{ij} = e^{-2\Psi}\delta_{ij}$, so

$$\Phi_0 = \ln \frac{1 - m_g/(2\varrho)}{1 + m_g/(2\varrho)}, \qquad \Psi_0 = -2\ln(1 + m_g/(2\varrho)). \tag{47}$$

These are physical log lapse and spatial conformal potential, equal to the Newtonian potentials at leading order. They are not the cosmological ADM auxiliary coordinates. Use the same real Dirichlet profile in all three columns,

$$f(\varrho) = \frac{\ell}{\sqrt{2\pi}\varrho} \sin \frac{\pi(\varrho - r_0)}{\ell}, \qquad \ell^{-3} \int_{r_0}^{L} f^2 \mathrm{d}^3x = 1, \quad \int |\nabla f|^2 \mathrm{d}^3x = \pi^2 \ell. \tag{48}$$

The variation is $\delta\Phi = q_A f$, $\delta\Psi = q_\psi f$, $\delta\eta = q_\varphi f T\eta$. Thus the inverse-metric columns are $\delta g^{00} = 2N^{-2} f q_A/c^2$ and $\delta g^{ij} = 2e^{2\Psi_0} f \delta^{ij} q_\psi$ in a time coordinate with $g_{00} = -N^2c^2$. The boundary variations vanish and the Gibbons–Hawking–York term implements fixed induced metric.

All Hessian entries below use the *static energy*, the negative of the static Lorentzian action per unit coordinate time. The exact Einstein energy restricted to this static conformal domain is

$$E_g = K \int e^{\Phi - \Psi}[2\nabla\Phi \cdot \nabla\Psi - |\nabla\Psi|^2]\mathrm{d}^3x, \qquad K = \frac{c^4}{8\pi G}. \tag{49}$$

To make the evaluated second variation explicit, set $E = e^{\Phi_0 - \Psi_0}$, $p = \nabla\Phi_0$, $q = \nabla\Psi_0$, $g = \nabla f$ and $B_0 = q^2 - 2p \cdot q$. Then

$$a/K = -\int E[f^2 B_0 - 4fq \cdot g]\mathrm{d}^3x, \tag{50}$$

$$b/K = -\int E[-f^2 B_0 + 4fq \cdot g - 2fp \cdot g - 2g^2]\mathrm{d}^3x, \tag{51}$$

$$c/K = -\int E[f^2 B_0 - 4fq \cdot g + 4fp \cdot g + 2g^2]\mathrm{d}^3x. \tag{52}$$

In vacuum the lapse constraint gives $a = 0$ after integration. The flat-limit check is $(a, b, c)/(K\ell) = (0, 2\pi^2, -2\pi^2)$, with $\det A_g/(K\ell)^2 = -4\pi^4$. The receipt evaluates the curved integrals and checks

their Hessian against independent finite differences. For this body's exterior the evaluated entries are

$$(a,b,c)/(KR) = (-1.27\times10^{-21},\ 19.739240289,\ -19.739240289), \tag{53}$$

$$(u,v) = (0,0), \qquad c_\varphi/(w_0\kappa_\eta R) = 9.86962014448, \tag{54}$$

$$(ac-b^2)/(KR)^2 = -389.637607185. \tag{55}$$

The profile norm is $1+2.2\times10^{-16}$, the metric condition number is 2.61803, the original annulus metric Euler residual is below 3.3×10^{-21} in R units and the projected inverse-response residual is zero. The near-zero a is below its quadrature error; the nonzero determinant comes from the mixed metric entry b. Indeed the vacuum equations and zero boundary traces give the exact simplification

$$A_g = 2KI_E\begin{pmatrix}0 & 1\\ 1 & -1\end{pmatrix}, \qquad I_E = \int E|\nabla f|^2 \mathrm{d}^3x, \tag{56}$$

which the unsimplified quadratures independently reproduce. Invertibility asserted here is that of this specified two-mode projected block. It is not an inverse for an unrestricted gravitational functional. Its indefinite sign is a static constraint property, not a propagating ghost test.

Since $j = A = F = 0$, Eq. (28) gives $u = v = 0$ pointwise; all remaining target metric derivatives vanish as well. The nonconstant phase self entry is nevertheless

$$c_\varphi = w_0\kappa_\eta \int E|\nabla f|^2 \mathrm{d}^3x > 0, \qquad k_{\mathrm{loc}} = c_\varphi, \qquad \mathcal{R}^{\mathrm{loc}}_{g\varphi} = 0. \tag{57}$$

Metric entries have energy units $K\ell$; phase entries have units $w_0\kappa_\eta\ell$ in the retained target action. Canonical phase $\phi_c = \sqrt{w_0\kappa_\eta}\,\delta q$ removes this unfixed coupling from the quadratic coefficient without calibrating it. Independently, the original Lorentzian Legendre matrix at $D\eta = 0$ is $w_0\kappa_\eta$ times the positive target metric (with the appropriate lapse/volume factors). Thus the physical phase kinetic condition also holds; it is not inferred from the gravitational static Hessian.

For an arbitrary small meridian phase ϕ, the actual linear source equation, with all quantities on the same annulus, is

$$\mathcal{L}_\phi\phi = -\partial_i(N\sqrt{h}\, w_0\kappa_\eta h^{ij}\partial_j\phi) = J_{\phi,\mathrm{grav}} + \mathcal{E}_\Sigma[Q_\phi] + \xi_\phi = 0, \quad \phi|_{r_0,L} = 0. \tag{58}$$

The vanishing gravitational source follows from the first target variation at $D\eta = 0$; the ordinary fluid action has no target-phase coupling in the declared sector. There is no internal target interface or random forcing in the specified problem. The conormal flux $N\sqrt{h}\, w_0(\kappa_\eta + X^3)h^{ij}\partial_j q$ vanishes on the reference, so the target-phase interface source is zero on both oriented boundaries. This does not set the separately exported topographic Q_Σ. Multiplication by ϕ and integration prove uniqueness of $\phi = 0$. This is an evaluated homogeneous source problem, rather than assigning zero to an unavailable source array.

The exported scalar ψ in Eq. (23) is not identified with this target phase by name. The derived map here is $\phi_c \mapsto \delta\eta = \phi_c T\eta/\sqrt{w_0\kappa_\eta}$ and its linear physical metric response is zero. A general constitutive map $\psi \mapsto \phi_c$, or extra cap/interface forcing of either field, is still required away from this target-phase problem. The solved zero phase does not determine the separately exported scalar or its source. No moving-seam law or statistical-state kernel is used.

9.4 Encounter, intrinsic derivatives and distinguishability

The nonzero local metric is the solved fluid interior and matched Schwarzschild exterior. At leading post-Newtonian order,

$$\Phi_N(r) = \Psi_N(r) = -\frac{GM}{Rc^2}\begin{cases} 15/8 - 5x^2/4 + 3x^4/8, & x \le 1, \\ 1/x, & x > 1. \end{cases} \tag{59}$$

The weak-field line-of-sight integral of $\Phi_N + \Psi_N$ gives a thin potential. For physical impact $b_\perp = |x_\perp|$, define $y = b_\perp/R$,

$$F_2(y) = \begin{cases} 1 - (1-y^2)^{5/2}, & 0 \le y < 1, \\ 1, & y \ge 1, \end{cases} \qquad \widehat{\alpha}(x_\perp) = \frac{4GM}{c^2 b_\perp} F_2(y) \frac{x_\perp}{b_\perp}. \tag{60}$$

The potential is the radial primitive of this expression; its additive constant cancels from relative delays. The finite support defines the same mass aperture in both models. The reduction retains leading post-Newtonian and thin-plane accuracy; the fractional omitted scales are $GM/(Rc^2)$ and R/D_{ab}. The exact TOV solution is not represented as an all-orders thin-plane optical calculation.

The parameter derivatives at fixed physical impact are obtained from this potential and its gradient, not a change in propagation maps. In particular $\partial_{\ln M}\widehat{\psi} = \widehat{\psi}$ in a mass-independent potential gauge and $\partial_{\ln M}\widehat{\alpha} = \widehat{\alpha}$. The radius derivative differentiates the same finite profile; all q derivatives vanish. In the complete image response these enter as

$$\delta\theta = J^{-1}\{\delta\beta + \delta G\widehat{\alpha} + G\widehat{H}\delta D_{ol}\theta + G\delta\widehat{\alpha}|_x\}, \qquad G = D_{os}^{-1}D_{ls}, \tag{61}$$

and the intrinsic arrival term is $-(1+z_l)\delta\widehat{\psi}|_x/c$ in the units of Eq. (20). Both terms are now implemented. The local body *replaces* the galaxy encounter for this comparison; it is not added to the SIS, and its potential is excluded from all three smooth propagation maps. The same fixed physical source, incoming-light preparation and observer clock protocol are retained.

Quantity	Evaluated conditional finite reference
Finite mass / areal radius	$10^{11}M_\odot$ / 3 kpc
$GM/(Rc^2)$	1.5951386e-06
Independent TOV residual (scaled)	4.27e-13
Maximum c_s^2/c^2	1.9939296e-06
Surface pressure / lapse-log residual	0 / 0
Image x positions [arcsec]	0.8354760, -0.0821619, -0.6574955
Relative delays [days]	21.7160766, 14.2714524
$\partial_{\ln M}\Delta t$ [days]	31.5546083, 7.1016644
$\partial_{\ln R}\Delta t$ [days]	-28.4907205, 0.0000000
Maximum image-equation residual [rad]	8.47e-22
Cosmological fixed/free-nuisance rank	3 / 0
Local q_0 response / free-nuisance rank	0 / 0

The receipt also gives mass/radius image and delay columns, finite-change checks, covariance, nuisance columns and singular values for this new family. The envelope and conventional predictions agree for every constant q and the same matter data: $\partial_q\theta = \partial_q\Delta t = 0$. Thus q is unidentifiable even with galaxy nuisances fixed. Nonzero mass/radius responses measure the shared body, not an additional BHSM effect. The six smooth-state columns are separately tested against this finite

body's source, mass, radius and mass-sheet nuisances. They have fixed-nuisance rank three and unrestricted rank zero; the separately derived finite-profile factorization has relative whitened residual 1.32×10^{-16}. No independent kinematic likelihood is assumed. This equivalence is the outcome for the declared source-free sector; it does not establish an equivalence theorem for charged interfaces or other reference states.

10 Scope of the derived and conditional results

The action-derived results are the target regularity/zero-flux theorem, the exact reference target equations, the solved finite-body Einstein and conservation equations, the matched exterior, the common-domain physical Hessian and the positive phase source problem. The weak-field optical integral then gives a nonzero conventional-equivalent encounter and evaluated intrinsic derivatives. The previous scalar-map nuisance null now has an analytic explanation, independently checked on its saved arrays.

Specified data are the target boundary value, positive target coupling, finite matter mass/radius and density law, source position, incoming light, axial cosmological harmonic and illustrative measurement covariance. The choice of retained effective gravity sector and absent extra cap reactions is conditional. The constant target state is admissible in that sector; no unique full-framework galaxy selector or nonzero BHSM correction is claimed. Its zero optical state derivative is a computed result.

The unresolved dependency is a general common-action map from exported topographic forcing to the physical target phase, including a nonzero interface/cap reaction if such data are prescribed. Equation (58) supplies a normalized target-phase operator; it does not replace the separate operator and source in Eq. (23). A stochastic covariance is unnecessary for the deterministic problem evaluated here. Additional independent physical information would be needed to identify state directions that remain degenerate with galaxy and source changes.

The reproducible commands are the original comparison script, the saved-array degeneracy check and the new local-family script; their exact arguments, pinned inputs, residuals and test receipts accompany the checkpoint. Baseline arrays remain unchanged. The numerical equivalence and rank-zero conclusions are limited to the declared families and approximations; they do not supply an observational detection forecast.

References

[1] P. Fleury, J. Larena and J.-P. Uzan, "Gravitational lenses in arbitrary space-times," *Classical and Quantum Gravity* **38** (2021), 085002. doi:10.1088/1361-6382/abea2d; arXiv:2011.04440v4.

[2] P. Fleury, J. Larena and J.-P. Uzan, "Line-of-sight effects in strong gravitational lensing," *Journal of Cosmology and Astroparticle Physics* **08** (2021), 024. doi:10.1088/1475-7516/2021/08/024; arXiv:2104.08883v5.

[3] L. Teodori, K. Blum, E. Castorina, M. Simonović and Y. Soreq, "Comments on the mass sheet degeneracy in cosmography analyses," *Journal of Cosmology and Astroparticle Physics* **07** (2022), 027. doi:10.1088/1475-7516/2022/07/027; arXiv:2201.05111v3.

[4] R. C. Tolman, "Static solutions of Einstein's field equations for spheres of fluid," *Physical Review* **55** (1939), 364–373. doi:10.1103/PhysRev.55.364.